

A novel classification model based on DWT and CNN-LSTM motor EEG imagination signals

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Abstract—Feature extraction and pattern classification play important roles in brain-computer interface (BCI) systems, which are widely used in rehabilitation medicine, artificial intelligence and other fields. Although advanced EEG acquisition technology can generate a large amount of EEG data for different brain regions, it inevitably has disadvantages such as high cost, long time consuming and inherent high false positives. Considering the contradiction between data volume and accuracy, this paper proposes a MI EEG feature extraction and classification method based on discrete wavelet transform (DWT) and Convolutional Neural Network-Long Short Term Memory (CNN-LSTM). Based on the project BCI motor imagination EEG(MI-EGG) data set, all evaluation indexes of the proposed CNN-LSTM model are superior to other models. The prediction accuracy, precision, recall rate, F1-score and MCC of C3 and C4 channels reaches 96.43%, 96.57%, 96.57%, 96.57% and 93.02%, respectively. The reliability of the proposed model is also proved by tenfold cross-validation. The average accuracy is 99.64%, which is far higher than the other models. It can be concluded that it is a reliable method to predict MI-EEG signals by using CNN-LSTM to train the modeling ability of images.

Keywords—Motion Imaging Electroencephalogram (MI EEG), Discrete Wavelet Transform (DWT), Convolutional Neural Networks (CNN), Long Short Term Memory (LSTM), Brain computer Interface (BCI)

I. INTRODUCTION

The development of brain computer interface (BCI) is a very promising research direction in the field of life science. Most non-invasive BCI collected the electroencephalogram (EEG) signals from the surface of the scalp [1]. Many fields of science and technology require high accuracy of EEG signals such as biomedicine, artificial intelligence and industry [2-4]. However, it is a very complex mission to process the signals because of the discrete, multidimensional and high-noise nature of the multichannel EEG [5]. The accuracy of identifying and classifying specific motion correlations in a single experimental mode was affected by these factors and thus required a large number of experimental measurements[6].

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Generally, EEG signals were first preprocessed by temporal or spatial filters such as bipolar filters [7], band pass filters [8], and common spatial patterns (CSP) [9-12]. Then a suitable feature extraction method was selected to process the preprocessed signal, which mainly contained logarithmic band power [13-15], autoregressive (AR) parameters [16-18] and time domain parameters [19-21] to reflect the characteristics of BCI. Finally, linear discriminant analysis (LDA) [22-26], support vector machines [27-30], neural networks [31-33] and nearest neighbor classifiers [34-36] were used to classify the characteristic samples.

Deep learning has recently reached state-of-the-art performance in several areas of pattern recognition [37]. It is suitable for nonlinear and non-stationary data. In addition, the extracted features can be analyzed to explain the decisions made by the network. For BCI applications, it provides encouraging results even in the absence of denoising algorithms. CNNs have been widely used in biomedical applications [38]. Li et al.[39] proposed an end-to-end CNN classifier consisting of nine layers for binary classification of normal and ictal EEG. In another study, Acharya et al.[40] introduced an end-to-end CNN classifier consisting of 13 layers for multi-class classification of EEG signals. In this study, they used EEG datasets from the University of Bonn [41], which include normal EEG, interictal EEG and ictal EEG. The researchers reported an overall accuracy of 88.67%. Ref. [42] proposed a CNN model and support vector machine system to replace the end-to-end CNN model with the classification accuracy of 86.25%. They used CNN to extract features from the amplitude spectrum of EEG signals and used SVM to classify EEG signals.

In the above studies, CNN was used as feature extraction, and deeper features were extracted by deepening the network model. Only spatial features were considered, but the time characteristics of EEG data model were not considered. To improve the recognition accuracy of BCI is one of the important tasks of establishing an effective and universal method for optimizing input data. In particular, the data processing efficiency is improved by reducing the complexity of the signal. To this end, a MI-EEG recognition or classification model is

established in this paper, including feature extraction (using DWT) and classification (using CNN-LSTM). Firstly, the Butterworth filter is used to obtain the EEG signals of C3 and C4 channels at 8~30Hz. Secondly, DWT is used to extract the time domain and frequency domain features of EEG signals, and the time-frequency feature map is generated. Thirdly, the features of the time-frequency feature map are successively input into the fixed-size image in channel order to generate the sample images required by CNN[43-47] and LSTM. Finally, the time-frequency images are trained in the CNN-LSTM model to obtain the classification results. The project BCI-EEG motor activity dataset and Dataset 1 of BCI competition IV are used to evaluate the performance of the proposed method in this study.

Compared with the existed research[48-50], our innovation can be summarized as follows: (1) only the data of C3 and C4 electrode channels are used, which improves the efficiency of data processing and facilitates the application in brain interface. (2) DWT is used to extract the time domain and frequency domain features of EEG, and the multi-channel EEG signals are converted into two-dimensional time-frequency images. (3) LSTM network is introduced to solve the problem of gradient explosion and gradient disappearance for EEG signal data with time characteristics. (4) Extensive experiments are conducted to our model and applied on dataset 1 of BCI competition IV, and excellent performance is achieved compared with other models.

II. MATERIALS AND METHODS

A. Data set and Data collection

In this study, we utilized the 1-1D motor data from the Project BCI MI-EEG dataset (<https://sites.google.com/site/projectbci/>), which documents the brain electrical activity of a 21-year-old right-handed male at different time intervals. These data were collected while the subject imagined random movements of the left and right hands with eyes closed. Each row in the dataset represents an electrode, including the C3 and C4 channels located in the brain's sensorimotor area. The signal recordings from these channels are crucial for understanding the brain's activity during motor imagery. The electrode order follows the layout of the international 10-20 system, including channels such as Fp1, Fp2, F3, F4, C3, C4, P3, P4, O1, O2, F7, F8, T3, T4, T5, T6, Fz, Cz, Pz, etc., with specific layout and labeling methods detailed in other literature [28]. Specifically, the C3 and C4 channels are located in the central region of the brain, corresponding to the imagery of left and right hand movements, and previous studies have shown strong signal responses in the surrounding sensorimotor cortex areas when imagining movements of the right and left hands [26].

In this study, we focus on analyzing Alpha and Beta wave signals, which are closely related to motor imagery. Through a review of existing literature, we found that using only C3 and C4 channels yields higher classification accuracy compared to using C3, C4, and CZ channels. Additionally, C3 and C4 channels already encompass the majority of effective information components [51]. Therefore, to enhance data processing efficiency and reduce computational complexity, we opt to concentrate on C3 and C4 channels.

Furthermore, in order to extract features related to motor imagery from these channels, we applied Butterworth bandpass filters to the signals for more accurately capturing signal components within the 8-30Hz frequency range. This filtering process aids in noise removal while preserving relevant signals, laying the groundwork for subsequent feature extraction and classification model development.

B. Discrete Wavelet Transform (DWT)

DWT is a widely used analysis method to extract EEG signal characteristics [52]. The advantage of wavelet transform is that the time domain and frequency domain can be extracted simultaneously. EEG signal is a low-frequency signal, with the main frequency below 30Hz, but it may be mixed with some high-frequency interference. In DWT algorithm, the performance of wavelet transform mainly depends on the choice of the mother wavelet $\Psi(t)$, and the selection criteria include symmetry, orthogonality, vanishing moment, compact support and fast algorithm [31], and $\Psi(t)$ can be defined as equation (1).

$$\Psi_{m,n}(t) = \frac{1}{\sqrt{|m|}} \Psi\left(\frac{t-n}{m}\right) \quad (1)$$

Where m is scale, n is a translation parameter.

There are two different function scales and wavelets used in DWT algorithm, and high and low-pass filters are required to connect these functions. Therefore, the algorithm has two lower samplers and two filters per phase. In this paper, down sampling outputs, high and low-pass filters are used to identify the details of EEG signals [11-14, 53]. After feature extraction of EEG signals through DWT, the signal correlation increases and the dimension decreases [34-35]. The parameters involved in the feature extraction process are as follows.

- (1) The mean of the sub-band coefficients

$$Mean = \frac{\sum_{i=1}^k c_i}{k} \quad (2)$$

- (2) The average power of sub-band wavelet coefficients

$$Average = \frac{1}{K} \sum_{i=0}^K (c_i)^2 \quad (3)$$

- (3) The standard deviation of the sub-band coefficient

$$Standard\ deviation = \sqrt{\frac{\sum_{i=1}^k (c_i - \mu)^2}{k}} \quad (4)$$

- (4) The ratio of the mean values of neighbor sub-bands

$$Ratio = \frac{\frac{\sum_{i=1}^k c_i}{k}}{\frac{\sum_{j=1}^k c_j}{k}} \quad (5)$$

During the experiment, different mother wavelet functions are tested and the optimal precision based on Daubechies 4 wavelet is obtained. The most accurate classification result can be obtained by extracting the feature through hierarchical

decomposition. The extracted time-frequency feature pictures are saved as training sample set.

C. Convolutional Neural Network - Long Short Term Memory

The time-frequency images after DWT are used as the training sample of CNNs. This paper adopts the shallow structure of a layer of convolutional network, as shown in Fig. 1. In Fig. 1, Convolution layer Conv 2D is used to extract the features of time-frequency images. The output of Conv 2D is used as the input of Pooling layer. The Pooling layer is used to further reduce the dimension of the information extracted by Conv 2D and the amount of calculation is reduced. The data of the convolution layer needs to be flattened before it can be directly added to the fully connected layer (FC), that is, the data of (height, width, channel) is compressed into one dimensional array of height $\times$ width $\times$ channels. The one-dimensional data obtained after flatten passes through FC and LSTM layers. The main function of the FC is to map the feature space calculated by the previous layer (convolution, pooling, etc.) to the sample label space. In view of the lack of long-term information storage in EEG data, LSTM is selected to selectively store and forget some information. Finally, it is classified by Softmax. In the proposed CNN-LSTM model, the number of convolution kernel is even. For shallow convolutional networks with small data sets, the kernel size should be adjusted according to the accuracy of classification.

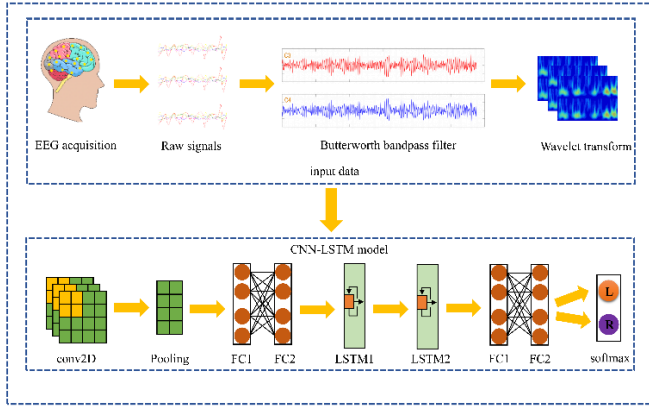


Fig. 1. Structure of the convolutional network-Long Short Term Memory.

(1) Convolutional layer

It consists of a convolution kernel that slides on an EEG time-frequency image. The kernel is the matrix convoluted with the input EEG time-frequency image. The stride controls the convolution of the filter with the input EEG time-frequency image. The output of the convolutional layer is also known as the feature map. The convolution operation is shown in equation (6).

$$b_i = \sum_{n=0}^{N-1} a_n b_{i-n} \quad (6)$$

Where a is time-frequency image, h is convolution kernel function, and N is the number of elements in a . The output feature map is b . The subscripts represent the n -th element of the feature map.

(2) Pooling layer

Pooling layer is the down-sampling layer. The function of pooling is to reduce the dimension of convolutional layer output neurons, and it can reduce the steps of operation and prevent the overfitting. In order to prevent the over-fitting of sample data and improve the generalization ability of the model when training the neural network model, dropout layer is added after each pooling layer to regularize the network. Dropout was more efficient than other standard regularization methods with less computational overhead, such as sparsely activated regularization, filter generic constraint, and weight attenuation[54]. The dropout operation is as follows equation(7) ~ equation(10).

$$r_j^{(l)} \sim \text{Bernoulli}(p) \quad (7)$$

$$\tilde{b}^{(l)} = r^{(l)} * c^{(l)} \quad (8)$$

$$z_i^{(l+1)} = w_i^{(l+1)} \tilde{b}^{(l)} + c_i^{(l+1)} \quad (9)$$

$$b_i^{(l+1)} = f(z_i^{(l+1)}) \quad (10)$$

$\text{Bernoulli}(p)$ is Bernoulli probability, which randomly generates mask vector r distributed between 0 and 1. The only element in r equals 0 and 1, and p is the probability of 1, and $(1-p)$ is the probability of 0. The dimension of r is the dimension of the input neuron in one layer.

In each mini-batch training, neurons set to 0 were randomly discarded according to the Bernoulli probability distribution. The mask vector is multiplied by the neurons corresponding to the layer. If the dimension of the mask vector is the same as that of the input neuron, the element is assigned a value of 0; otherwise, it is assigned a value of 1. In order to prevent the large network from being troublesome and overfitting, dropout randomly selected some neurons to discard temporarily and found a thinner network from the original network. Assuming that the dropout ratio is p , then the probability of each output node setting 0 is p , and these neurons do not participate in training, and their weights are not updated. Only the corresponding parameters of the remaining neurons can be updated by back propagation. The Dropout layer processes all neurons directly forward during the test. The complete process is shown in Fig. 2.

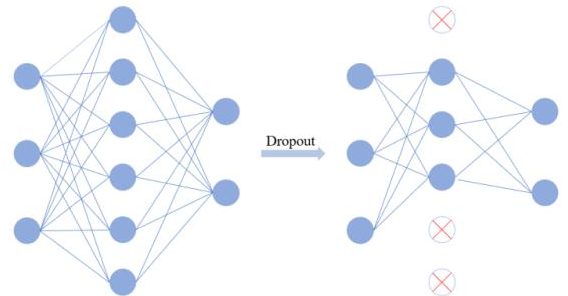


Fig. 2. Dropout layer processing.

(3) Fully connected layer

This layer fully connects all activation neurons of the previous layer. There are two activation function in this structure,

that is, Rectified linear activation unit (LeakyRelu) and Softmax. LeakyRelu maps the input of neurons to the output after every convolutional layer, widely used in the field of deep learning [55]. Activation function is introduced to increase the nonlinearity of neural network model. Its output has a slight slope against the negative input in the network structure. Therefore, it can solve the problem that the neuron does not learn when the Relu function enters the negative interval. Leaky helps expand the range of ReLu functions, which typically have a value of a around 0.01. The function LeakyRelu shows as equation (11).

$$f(a) = \begin{cases} a & \text{if } a > 0 \\ 0.01a & \text{otherwise} \end{cases} \quad (11)$$

(4) Softmax

Softmax is often used to calculate the probability distribution of C class regression. z_i is the output value of the i -th node, and C is the number of nodes, that is, the number of possible output classification categories. The Softmax function can convert the output value of the multi-classification problem into a probability distribution in the range $[0,1]$ and the sum of all classification results is 1. In this paper, the Softmax function is used at the fully connected layer to predict whether the input EEG time-frequency image is a left or right-hand motion signal.

$$\text{Softmax}(z_i) = \frac{e^{z_i}}{\sum_{c=0}^C e^{z_c}} \quad (12)$$

(5) Long Short Term Memory unit(LSTM)

LSTM is a special type of recursive neural network (RNN), which can learn long-term dependency information and be suitable for applications such as natural language processing and biomedical science. The unit structure of LSTM is shown in Fig. 3.

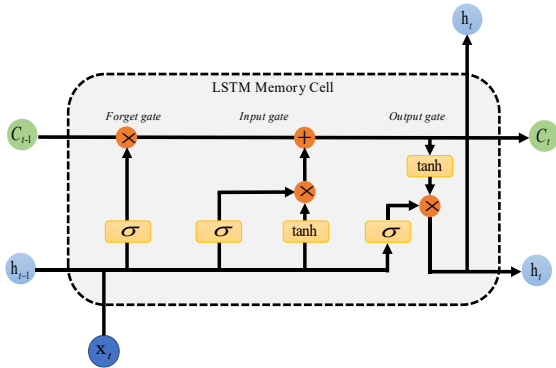


Fig. 3. Long Short Term Memory neural cell.

In Fig.3, there are four gates in the LSTM block. The unit state gate Z remembers information over time. The forget gate Z_f controls the range of values stored in the unit. The input gate Z_i controls the range of value flow in the unit. The range of values can be used to calculate the output in the output gate Z_o control unit. Each gate contains a full connectivity layer and activation capabilities. In addition, there are three inputs, that is, cell state C_{t-1} , hidden state h_{t-1} and current input x_t . The corresponding outputs are state C_t , hidden state h_t , and output

y_t . The current output is generated based on the hidden state. LSTM controls the transmission state through the gate state to remember the information that needs to be remembered for a long time and forget the unimportant information. Compared with RNN with only one memory overlay mode, LSTM is particularly useful for many tasks requiring long-term memory. The mathematical expressions of LSTM units are defined as equation(13) ~ equation(19).

$$Z_f = \sigma(W_f[x_t, h_{t-1}]) \quad (13)$$

$$Z_i = \sigma(W_i[x_t, h_{t-1}]) \quad (14)$$

$$Z = \sigma(W[x_t, h_{t-1}]) \quad (15)$$

$$Z_o = \sigma(W_o[x_t, h_{t-1}]) \quad (16)$$

$$C_t = Z_f \times C_{t-1} + Z_i \times Z \quad (17)$$

$$h_t = Z_o \times \tanh(C_t) \quad (18)$$

$$y^t = W' h_t \quad (19)$$

D. Method Flowchart

The method flowchart, as shown in Fig. 4, clearly illustrates the steps for processing and classifying MI-EEG signals using the proposed CNN-LSTM model.

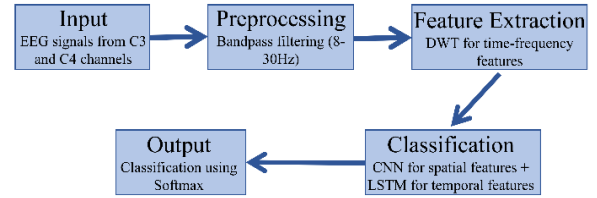


Fig. 4. Method Flowchart.

E. Evaluation indexes

Now assume that there are only two categories of classification targets, namely positive cases and negative cases respectively. There are five evaluation indexes shown as equation(20) ~ equation(24).

$$\text{precision} = \frac{TP}{TP+FP} \quad (20)$$

$$\text{recall} = \frac{TP}{TP+FN} \quad (21)$$

$$\text{accuracy} = \frac{\text{right}}{\text{all}} \quad (22)$$

$$F1 - \text{score} = \frac{2}{\frac{1}{\text{precision}} + \frac{1}{\text{recall}}} \quad (23)$$

$$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP+FP)(TP+FN)(TN+FP)(TN+FN)}} \quad (24)$$

Where, True positives (TP) is the number of cases that were actually positive and were classified as positive by the classifier.

False positives (FP) is the number of false positive cases that were wrongly classified as positive cases, i.e. the number of positive cases that were actually negative cases but were classified by the classifier. False negatives (FN) represent the number of negative cases that are incorrectly classified as negative, that is, the number of negatives that are actually positive cases but classified as negative by the classifier. True negatives (TN) represent the number of negative cases that are correctly classified as negative, that is, the number of negatives that are actually negative cases and are classified as negative by the classifier.

Precision is a measure of accuracy, representing the percentage of positive cases that are actually positive. Accuracy is the most common evaluation index. The number of correct samples is divided by the number of all samples. Generally speaking, the higher is the accuracy, the better is the classifier. Recall rate is a measure of coverage, which measures how many positive cases are divided into positive cases. F1-score is the harmonic average of accuracy and recall rate. Matthews correlation coefficient (MCC) is an indicator used to measure the classification performance of binary classification, which can be applied even when the samples of the two classes are not balanced. The range of MCC value is $[-1, 1]$, 1 indicates a perfect prediction of the subject, 0 indicates that the predicted result is not as good as the random prediction result, -1 refers to the predicted classification is completely inconsistent with the actual classification.

III. RESULTS AND DISCUSSION

The data set used by the CNN in this paper is 140 time-frequency images after Butterworth filter and DWT. They are compressed into 800×600 in size and stored in PNG format. Each picture is a time-frequency image composed of two channels of MI-EEG signal after DWT. Two time-frequency images are randomly selected for each category of left and right-handed imaginary movements. It is difficult to distinguish the characteristics of each category by human eyes. After DWT, the x-axis should be time/s, and the y-axis should be frequency/Hz. The motion of the left hand and the right hand can be reflected in the time-frequency image, which provides a premise for the feature extraction of the deep learning model.

In the training process of CNN-LSTM, time-frequency image is the input to train the classification model. 1 and 2 are used as labels for left and right-hand MI-EEG, respectively. At the same time, datasets are segmented in the experiment, 90% of which is training sets, 10% is verification sets, and the training times are 100, and each training set and test set were not re-split. The number of convolution kernels is set to 64, so the resulting feature map is 64. However, each filter in the convolution layer is different, so different feature graphs are obtained after the convolution operation, and the blank feature graphs are caused by the fact that the parameters on the filter are not suitable for extracting the time-frequency images of this experiment.

The proposed CNN-LSTM model is verified on the data sets that do not appear in the training set. In order to verify the performance of the model proposed in this paper, the same data has been used on some classical models, such as CNN-LSTM

density network, mobile network, ResNet, VGG16, VGG19, XceptionCNN and DNN. Only LSTM structure is missing in CNN and DNN. Table I shows the performance of these methods, including accuracy, precision, recall, F1-Score and MCC.

TABLE I. THE PERFORMANCE OF CNN-LSTM COMPARED WITH OTHER MODELS

Model	Accuracy	Precision	Recall	F1-score	MCC
DenseNet	78.57%	82.14%	78.85%	79.00%	60.41%
MobileNet	89.29%	90.85%	89.28%	88.85%	79.47%
ResNet	42.86%	18.42%	42.85%	25.71%	0%
VGG16	53.57%	28.92%	53.57%	37.50%	0%
VGG19	53.57%	28.92%	53.57%	37.50%	0%
Xception	78.57%	79.57%	78.42%	78.71%	57.73%
CNN	92.86%	88.85%	93.14%	92.57%	86.60%
DNN	85.71%	86.42%	85.57%	86.00%	72.30%
CNN-LSTM	96.43%	96.57%	96.57%	96.57%	93.02%

From Table I, it can be seen that our method outperforms all other models in terms of precision (96.43%), precision (96.57%), recall (96.57%), F1-score (96.57%) and MCC (93.02%). Surprisingly, VGG16 and VGG19 performed the worst, with Accuracy (53.57%), precision (28.92%), recall (53.57%), F1-Score (37.50%) and MCC (30%), which may be due to the loss of image features after multiple convolutions. The same is true for DenseNet and ResNet because the depth of the convolutional layer is too long. For MobileNet and Xception, they are both lightweight networks, and their convolution model retains the extracted features well. So their performance is better than DenseNet and ResNet. DNN has no convolution operation, it cannot extract the features of EEG time-frequency images, so its performance is not very good. The accuracy of CNN-LSTM model proposed in this paper is 3.57% higher than that of CNN (92.86%). The precision of CNN-LSTM model proposed in this paper is 7.72% higher than that of CNN (88.85%). The recall rate of CNN-LSTM model proposed in this paper is 3.43% higher than that of CNN (93.14%). The F1-Score of CNN-LSTM model proposed in this paper is 4 % higher than that of CNN (95.57%). The MCC of CNN-LSTM model proposed in this paper is 6.42 % higher than that of CNN (86.6%). CNN has slightly worse performance than CNN-LSTM, which may be due to the small number of data sets. Overall, CNN-LSTM has best performance in our experiment. CNN can integrate shallow features to obtain deep abstract features. LSTM is a time series structure that can describe and predict the characteristics of EEG signals with good spatial-temporal correlation. Therefore, it is possible to find features that can effectively represent the distribution of the data training set. And the highest accuracy of the prediction results are 96.43%, respectively, which indicates that there is only one image classification error in the test set, and the experiment has achieved very good results.

The CNN-LSTM model proposed in this paper has achieved good results in the task of MI-EEG signal classification. In order to further verify the accuracy advantage and stability of CNN-LSTM model, ten-fold cross validation method is used and the results are compared with other models. Cross validation refers to the process of repeatedly selecting training sets and traversing all data for verification. We use 90% of the images for training.

Cross validation is used on 90% images to train and find the best hyperparameters. After the training is completed, the model is evaluated on the test set (10% of the images not seen by the model saved separately). The experimental results are shown in Table II.

TABLE II. COMPARISON OF CNN-LSTM AND OTHER MODELS THROUGH TEN-FOLD CROSS-VALIDATION

Methods	K1	K2	K3	K4	K5
DenseNet	92.86%	75.00%	89.29%	71.43%	67.86%
MobileNet	82.14%	89.29%	78.57%	78.57%	67.86%
ResNet	57.14%	53.57%	42.86%	60.71%	57.14%
VGG16	42.86%	46.43%	42.86%	60.71%	57.14%
VGG19	57.14%	46.43%	67.86%	60.71%	57.14%
Xception	92.86%	64.29%	92.86%	53.57%	75.00%
CNN	100%	100%	100%	96.43%	100%
DNN	57.14%	53.57%	96.43%	39.29%	100%
CNN-LSTM	100%	100%	96.43%	100%	100%
K6	K7	K8	K9	K10	Mean
57.14%	89.29%	82.14%	57.14%	57.14%	73.92%
92.86%	67.86%	85.71%	45.86%	60.71%	74.94%
50.00%	50.00%	35.71%	57.14%	57.14%	52.14%
50.00%	50.00%	35.71%	57.14%	57.14%	49.99%
50.00%	50.00%	64.29%	57.14%	57.14%	56.78%
85.71%	89.29%	64.29%	42.86%	57.14%	71.78%
100%	50%	96.43%	100%	96.43%	93.92%
50.00%	96.43%	96.43%	42.86%	96.43%	72.85%
100%	100%	100%	100%	100%	99.64%

From Table II, K1 to K10 are the accuracy of each model after each cross-validation, and finally we calculate the average accuracy of each model. It can be seen from the Table II that our model has the highest average accuracy of 99.64%, CNN is second only to CNN-LSTM with an average accuracy of 93.92%, and VGG16 has the worst average accuracy of 49.99%. The stability of the test performance can be guaranteed by ten-fold cross-validation, which shows that the CNN-LSTM model is effective for MI-EEG signal classification.

IV. CONCLUSIONS AND DISCUSSION

In recent years, a growing number of deep learning-based MI-EEG prediction studies have been carried out in the biomedical field. CNN and LSTM are one of the most common and effective binary classification processing methods for EEG signals. In addition to motor image-related EEG classification, it is also important for the classification of brain activity such as sleep apnea psychiatric disorders and seizure patterns.

In this paper, a binary classification method for EEG signals based on wavelet transform and convolutional neural network is proposed. The model is trained and tested using 1-1D motion data from the project BCI MI-EEG signal dataset. Since the wavelet transform preserves both the time-domain and frequency-domain characteristics of the signal, the time-frequency image generated by the wavelet transform is used as the input of the model. According to the original channel order, the image preservation form is arranged to reduce the coupling loss between data channels. The form of time-frequency image

is used as input, and the extract ability of convolutional network is used to provide a new means for EEG signal classification. For MI-EEG data, CNN extracts spatial features and LSTM extracts temporal features, and classifying MI-EEG signals. Using the modeling capabilities of CNN and LSTM is an innovative method to analyze and predict MI-EEG signals. The effectiveness of CNN-LSTM is verified by comparing with other EEG classification methods. Since CNN and LSTM can extract deep features, the accuracy of CNN-LSTM for MI-EEG signal classification can reach more than 99%. This CNN-LSTM model performs well on different evaluation metrics. The stability of the model is further verified by 10-fold cross validation, and the average accuracy reaches 99.28%, which is much higher than other models. The results show that the proposed model has good universality and robustness. Therefore, it can be considered that the proposed method can improve the accuracy of MI-EEG signal prediction and the efficiency of the related brain-computer interface.

Although good experimental results are achieved by using two-dimensional images as input in our proposed model, it still increases the calculation amount of model training compared with one-dimensional data. In the future, deep separable convolutional neural networks will be used, which save performance overhead compared to convolutional neural networks. Similarly, our proposed deep learning model can also be extended to applications such as epilepsy diagnosis by EEG classification. Further model improvements can be planned by integrating more advanced feature extraction techniques and deep learning algorithms to identify and enhance key EEG signal features that can capture pre-seizure epileptic activity, thereby improving prediction accuracy and real-time responsiveness. We believe that through continuous technological innovation and model optimization, this research will deliver more precise medical services and higher quality of life for patients.

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